

Ayush Agarwal

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Experience

Docu3C Technologies

AI/ML Intern

Reported to: Daniel Ingitaraj K, Head of Product Engineering

- Developed and deployed AI/LLM-based applications for financial document intelligence using Azure AI ecosystem
- Built end-to-end machine learning and LLM pipelines with structured outputs using Pydantic and schema validation
- Designed multi-agent systems and tool-calling workflows using MCP (Model Context Protocol)
- Implemented LLM orchestration pipelines using Instructor for structured generation
- Built scalable backend systems with FastAPI, supporting real-time streaming (SSE/WebSockets) and async execution
- Developed interactive UI applications (Streamlit/React) for AI-driven workflows and validation systems
- Leveraged Azure ML, Cognitive Services, and GPU infrastructure for scalable experimentation and deployment
- Maintained production-grade code quality using PEP8 standards, Ruff linting, and modular architecture
- Utilized modern tooling (uv) for efficient dependency and environment management
- Conducted benchmarking, evaluation, and iterative optimization of AI systems
- Collaborated in building scalable AI systems and participated in team-based engineering workflows

Academic Projects

Hybrid Legal RAG: Sparse-Dense-Graph Retrieval System

End-to-End AI System

- Engineered a production-grade retrieval-augmented system combining BM25 (FTSS), dense embeddings (MPNet + FAISS), and graph-based ranking (PageRank)
- Implemented Reciprocal Rank Fusion (RRF) to unify multiple retrieval modalities for improved ranking quality
- Processed and indexed 26K+ documents, enabling high-precision semantic and keyword-based search
- Achieved 82.4% Top-1 accuracy and NDCG@10 of 0.89, outperforming single-modality baselines
- Built a FastAPI backend supporting search, conversational querying, and structured summarization
- Conducted ablation studies to validate robustness under high-noise conditions
- Containerized deployment using Docker with memory-efficient optimizations

PolarBrief AI – Legal Argument Analyzer

Full-Stack LLM Application

- Developed an AI-powered document analysis tool to extract, classify, and score arguments from unstructured PDFs
- Integrated LLMs via LangChain for summarization, heading generation, and structured scoring
- Implemented argument polarity classification (Pro/Con/Neutral) with weighted ranking using TF-IDF + LLM outputs
- Designed a hybrid OCR + text extraction pipeline using Tesseract and PDF parsers
- Generated structured outputs in JSON/PDF formats with citation-aware summaries
- Built an interactive Streamlit interface for visualization and report generation

Knowledge Graph Pipeline for Document Intelligence

- Built an end-to-end knowledge graph extraction pipeline from unstructured documents using NLP and LLMs
- Implemented multi-stage processing: OCR, cleaning, chunking, entity-relation extraction, and validation
- Designed deterministic ID mapping and cross-chunk entity resolution using fuzzy matching
- Normalized entities and relations to construct a clean, canonical knowledge graph
- Visualized graphs using NetworkX and PyVis for interactive exploration
- Exported structured graph data for downstream analytics

WESAD Multimodal Emotion Recognition

- Implemented and compared ML models (LDA, Random Forest, Decision Tree, AdaBoost, KNN) for emotion classification
- Processed multimodal physiological signals (ECG, EDA, EMG, Respiration, Temperature, BVP, ACC)
- Designed experiments to evaluate sensor modality combinations (chest vs wrist vs combined)
- Built pipelines for cross-validation, accuracy, and F1-score evaluation
- Generated detailed reports identifying optimal modality configurations

Technical Skills

- **Languages:** Python, TypeScript
- **AI/ML:** Machine Learning, Deep Learning, NLP, LLMs, RAG
- **Libraries/Frameworks:** Scikit-learn, Pandas, NumPy, NLTK, spaCy, LangChain
- **Models/Techniques:** LDA, Random Forest, Decision Tree, AdaBoost, KNN, TF-IDF, NER, Embeddings (MPNet), Prompt Engineering

- **Retrieval/Graph:** FAISS, BM25, Knowledge Graphs, PageRank, RRF, Vector Databases
- **Backend:** FastAPI, REST APIs, WebSockets, SSE, Async Programming
- **Frontend:** React, Streamlit, Tailwind CSS
- **Databases:** SQLite (FTSS), Azure CosmosDB, Azure Blob Storage
- **Cloud/DevOps:** Microsoft Azure, Docker
- **AI Systems:** Multi-Agent Systems, Function Calling, MCP, Pydantic, Instructor
- **Tools:** Tesseract OCR, pdfplumber, pdf2image, PIL, NetworkX, PyVis, RapidFuzz, Joblib
- **Practices:** uv, Ruff, PEP8
- **Auth/Security:** OAuth 2.0, MSAL
- **Monitoring:** OpenTelemetry, Arie Phoenix

Education

Degree CGPA	Specialization	Tenure	Institution
B. Tech 8.52/10	CSE (Core)	Sep 2023 - Ongoing	VIT Bhopal University

Certifications

- Prompt Engineering for Developers - DeepLearning.AI (Andrew Ng)
- Machine Learning - NPTEL
- Introduction to Large Language Models - Google Cloud
- Generative AI - Google Cloud

Co-curriculars & Achievements

- **Coding:** Solved 125+ Data Structures & Algorithms problems on LeetCode
- **Achievements:** Worked on advanced AI/ML and LLM-based systems including RAG pipelines, multi-agent systems, and document intelligence. Gained hands-on experience with Azure AI ecosystem and production deployments. Actively engaged in collaborative development and agile workflows.
- **Responsibilities:** Event Lead, Linpack Club, VIT Bhopal University. Organized and managed technical events and workshops, coordinating with teams and handling end-to-end execution. Led planning and operations, ensuring smooth execution and participant engagement.
- **Extracurricular:** Volleyball (represented Thunders at Aavahan, VIT)

Additional Information

- **Hobbies:** Exploring AI advancements, Volleyball
- **Languages:** English, Hindi